

## Major Project Presentation

# **Rentmart : A Web-based Platform For Renting Construction Equipment**

Presented by

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**(Formerly West Bengal University of Technology)**

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# OBJECTIVE

- **To develop an online equipment rental marketplace that connects equipment owners with people who need equipment for short-term use.**
- **To support users different areas where access to equipment is limited and traditional rental methods are slow or inefficient.**
- **To make equipment easily searchable and accessible by allowing users to browse, filter, and compare different equipment based on category, price, and location.**
- **To provide a platform for equipment owners to list and rent their unused machines or tools, helping them earn additional income.**

# ABSTRACT

- **The platform provides features such as equipment listing, secure authentication, booking management, and reviews/ratings to build trust between users.**
- **The system mainly aims to help rural and local users access equipment easily while allowing owners to earn from unused resources.**
- **The project develops a web-based Equipment Rental Marketplace that connects equipment owners with people who need equipment for short-term use.**
- **The platform aims to reduce equipment costs, improve resource utilization, and support local and rural communities through easy access to shared equipment.**

# TECH STACK

## ▪ Frontend

- **Framework** : React.js , Next.js
- **Styling** : Tailwind CSS, Shadcn UI
- **Language** : Typescript
- **Tools** : React Query , Lucide Icon

## ▪ Backend

- **Framework** : Express.js
- **Supporting Libraries** : nodemailer, twilio, Winston logger, jwt

## ▪ Database

- **Database** : PostgreSQL
- **ORM** : Prisma

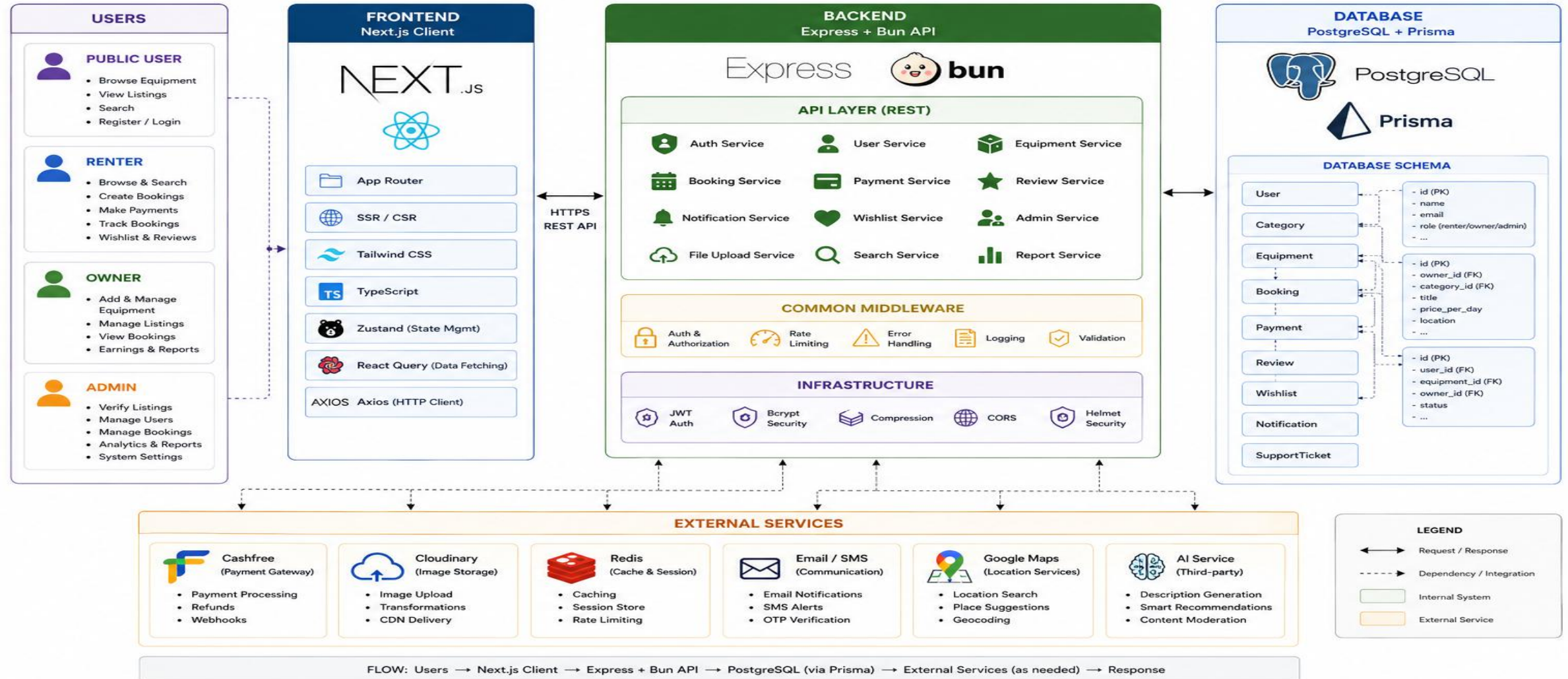
## ▪ External services

- Google maps
- Redis
- AI Service (Gemini)
- Cloudinary
- **Payment Gateway** : Cashfree

# ARCHITECTURE

## RENTMART – SYSTEM ARCHITECTURE

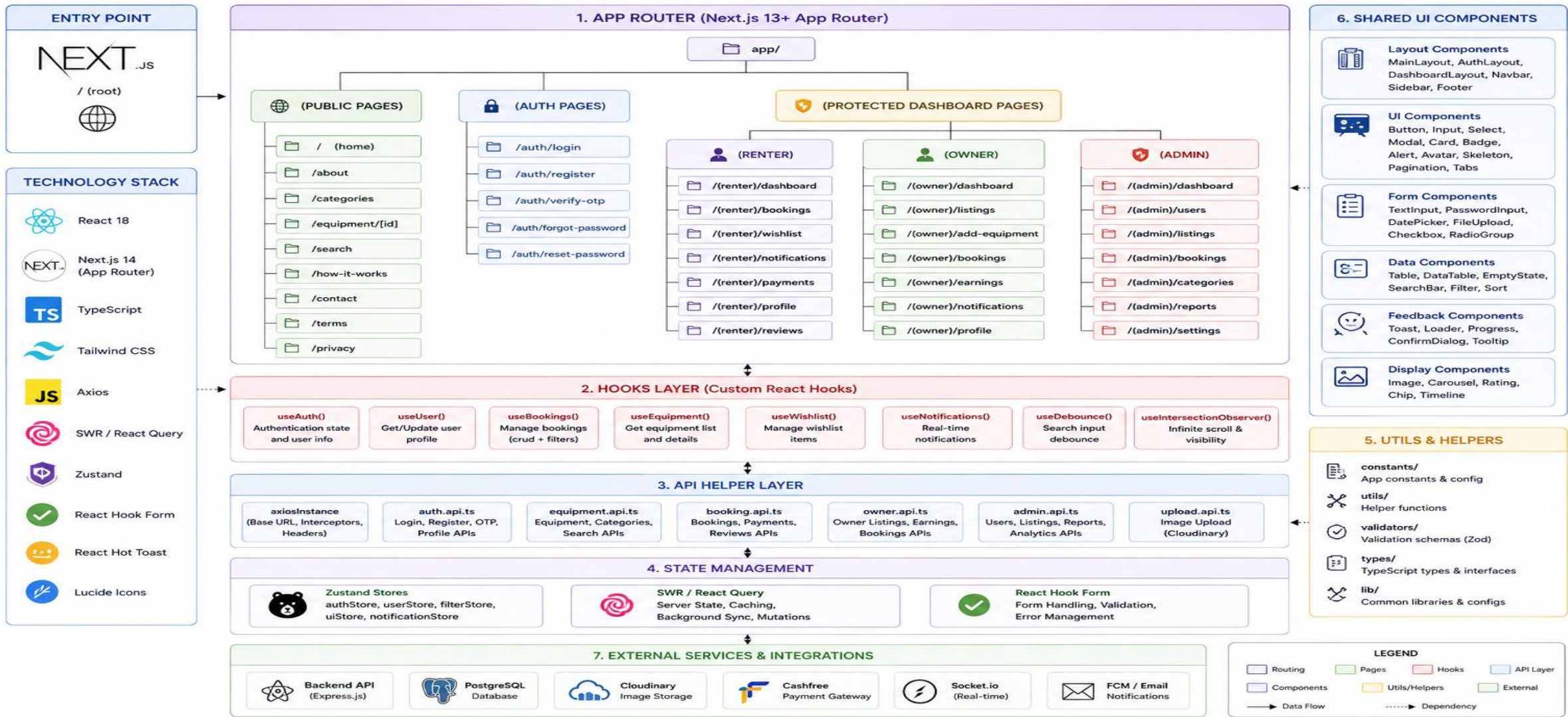
Equipment Rental Marketplace





# FRONTEND ARCHITECTURE

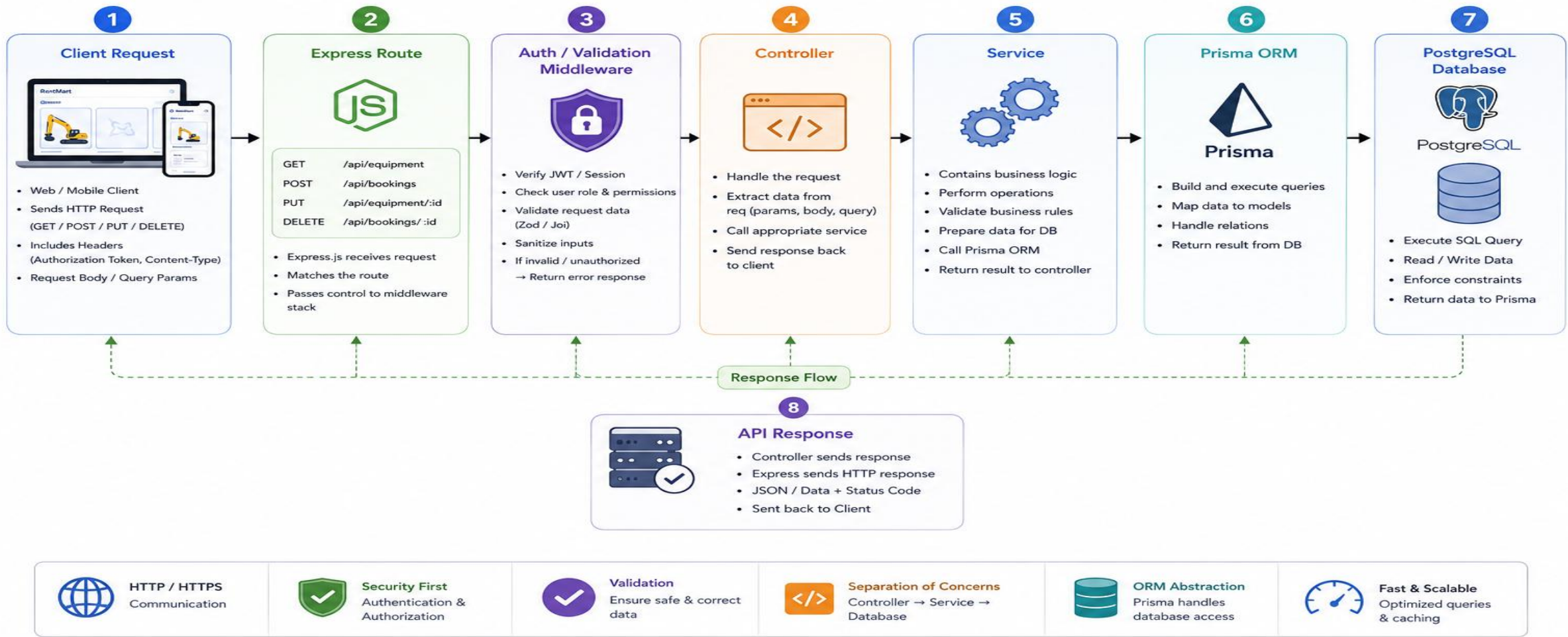
## RENTMART – FRONTEND ARCHITECTURE (NEXT.JS)



# BACKEND ARCHITECTURE

## RentMart – Backend Request Flow Diagram

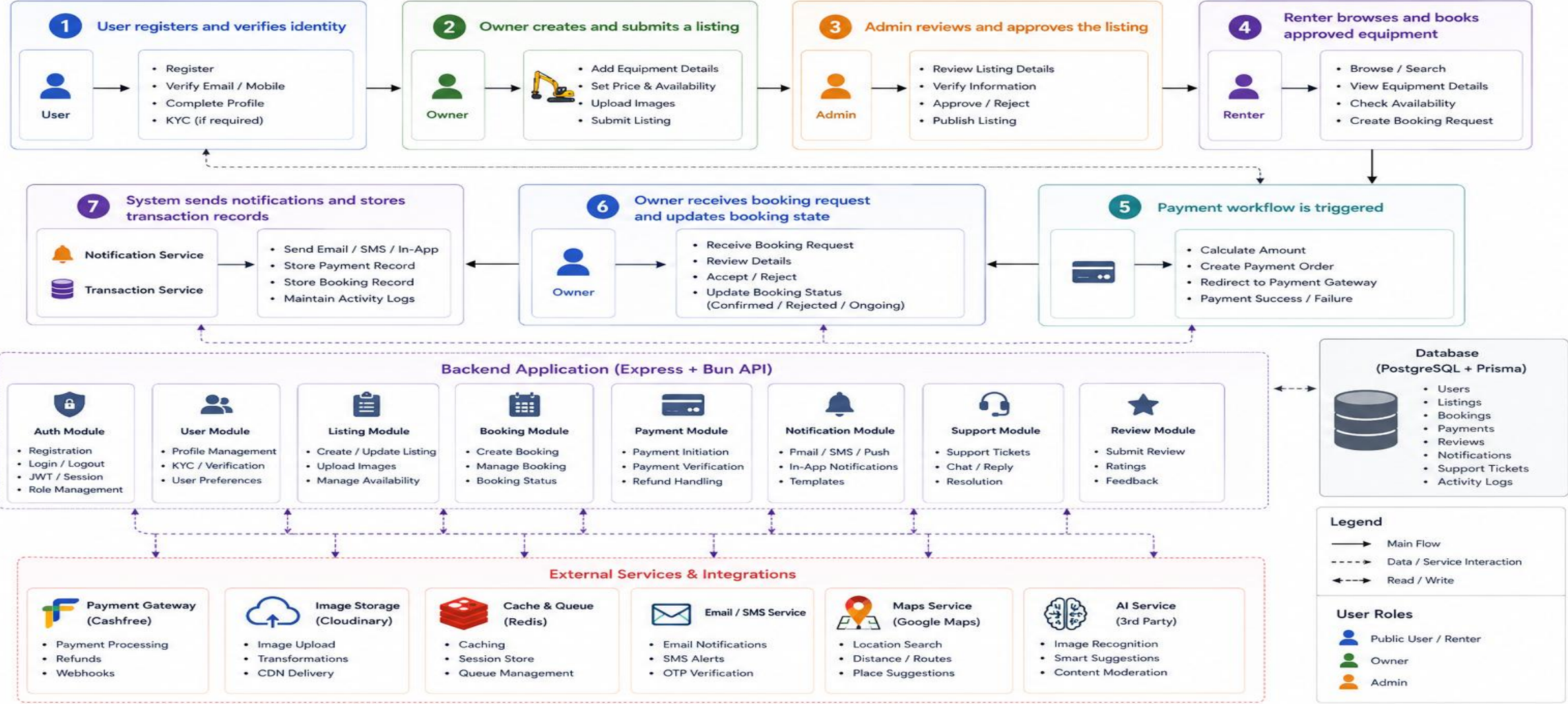
Overview of how a request travels from the client to the database and back





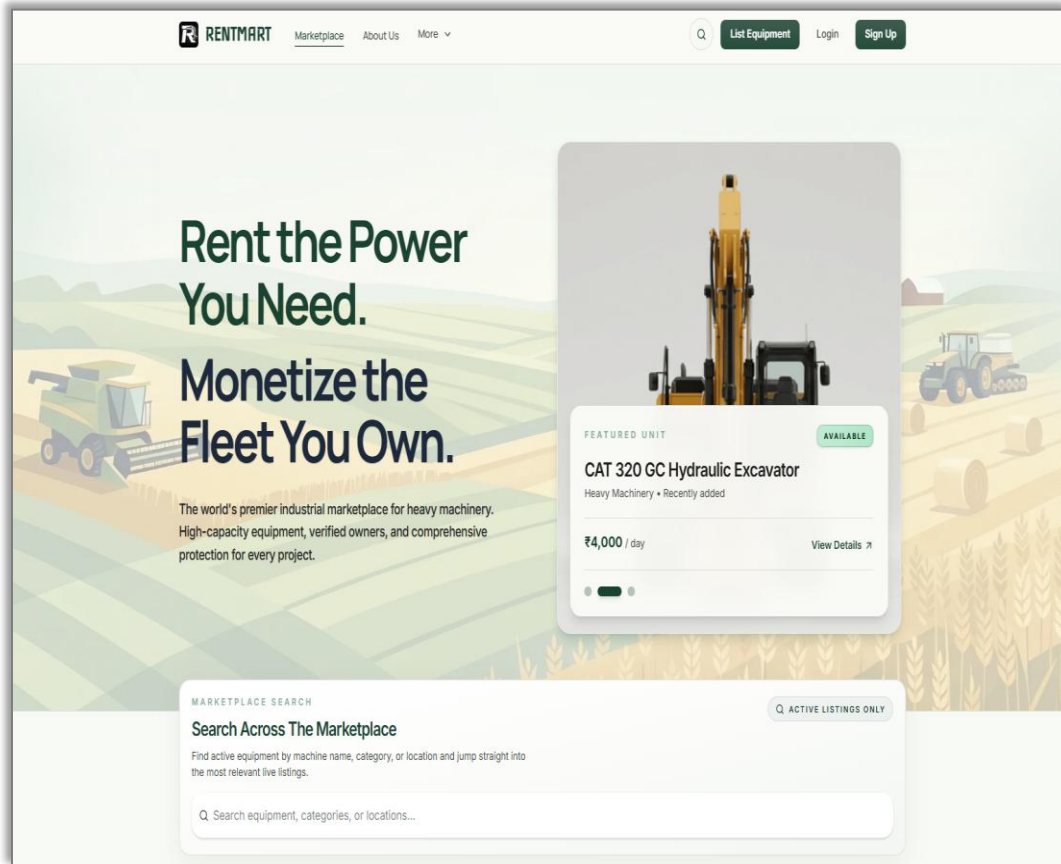
# WORKFLOW

This diagram illustrates how different modules interact with each other to complete a rental transaction flow.

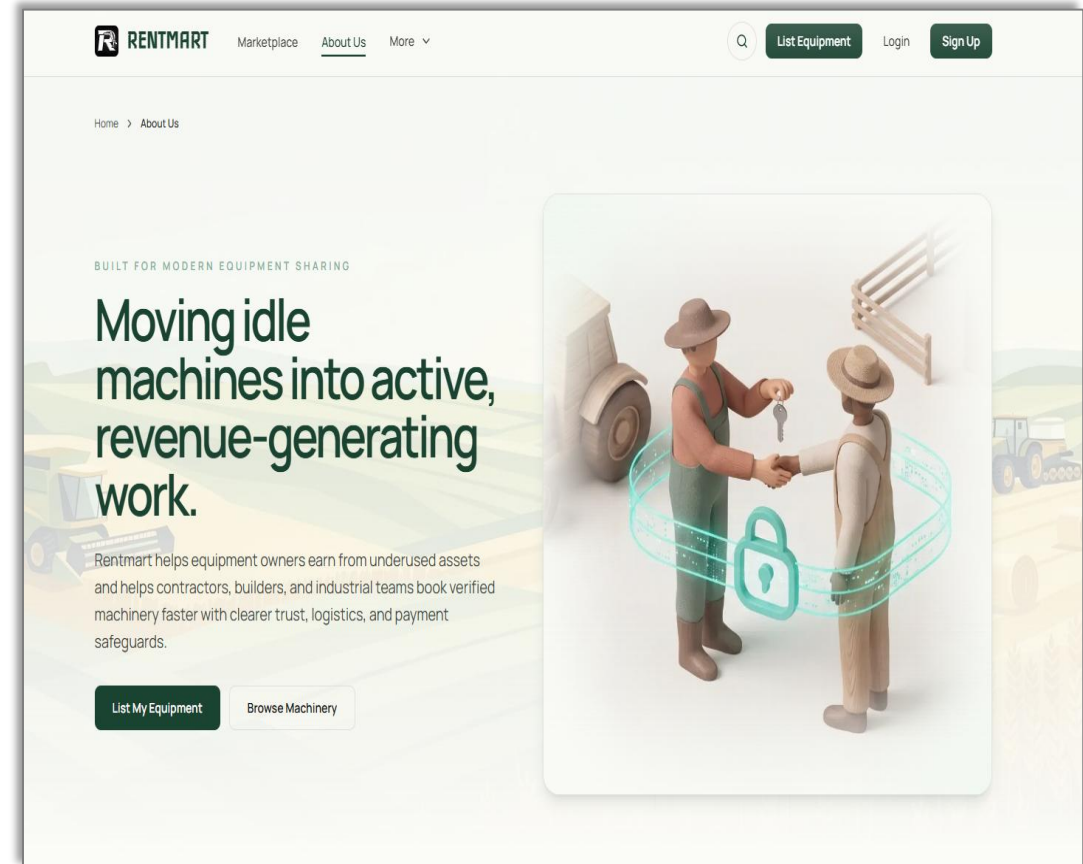




# LANDING PAGE OVERVIEW



*fig: Landing page*



*fig: About page*

# AUTHENTICATION PAGES

RENTMART

Marketplace

About Us

More ▾

List Equipment

Login

Sign Up

## Join the Future of Rural Machinery

Experience the most reliable machinery ecosystem. Precision in procurement for owners and renters alike.

 **SECURE PAYMENTS**  
Escrow-protected transactions for every rental contract.

 **VERIFIED USERS**  
Strict identity verification for a high-trust marketplace.

 **PLATFORM INSURANCE**  
Comprehensive damage coverage on all industrial equipment.

### Create your account

Join our industrial network and start trading today.

☐ I am an Owner  
List machinery and manage your fleet.

☒ I am a Renter  
Find and book industrial equipment.

FULL NAME

John Doe

EMAIL ADDRESS

john@industrial.com

BUSINESS ADDRESS

123 Industrial Way

PASSWORD

\*\*\*\*\*

CONFIRM PASSWORD

\*\*\*\*\*

Create Account

Already have an account? [Sign in.](#)

By signing up, you agree to our [Terms of Service](#) and [Privacy Policy](#).

© 2026 Rentmart Industrial. All rights reserved. Precision in Procurement.

fig: Sign-Up page

RENTMART

Marketplace

About Us

More ▾

List Equipment

Login

Sign Up

## Welcome back to Rentmart

Please enter your details to access your dashboard.

EMAIL ADDRESS

name@company.com

PASSWORD

\*\*\*\*\*

☐ Remember me

[Forgot Password?](#)

Sign In

Don't have an account? [Sign up.](#)



Over 5,000+

safe rentals completed.

Experience the most reliable machinery ecosystem for owners and renters alike.

fig: Sign-in page

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# PRODUCT OVERVIEW PAGES

RENTMART

Marketplace

About Us

More ▾

Q

List Equipment

Sign In

Sign Up

Home > Power Tools > VoltMax Pro 20V Cordless Kit







ACTIVE

RJ

Rahul Jain

Member since 1 May 2026

VoltMax Pro 20V Cordless Kit

₹700 / day

Escrow-Protected Booking

Rentmart only places the authorization hold after the owner approves your request. Your refundable security hold is 30% of the rental fee, capped between ₹500 and ₹5,000, and is only captured if damage is reported.

Rental Fee

Select dates

Platform Fee (10%)

Select dates

Damage Waiver

₹999

Refundable Security Hold (30%)

Select dates

Sign In to Book

Submit your date request first. If the owner approves, you will get a 1-hour window to complete payment authorization.

Booking Protection

Damage waiver and platform charges are mandatory. Payment authorization happens after owner approval, and final routing completes after safe return or dispute

fig: Product page

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# ADMIN DASHBOARD

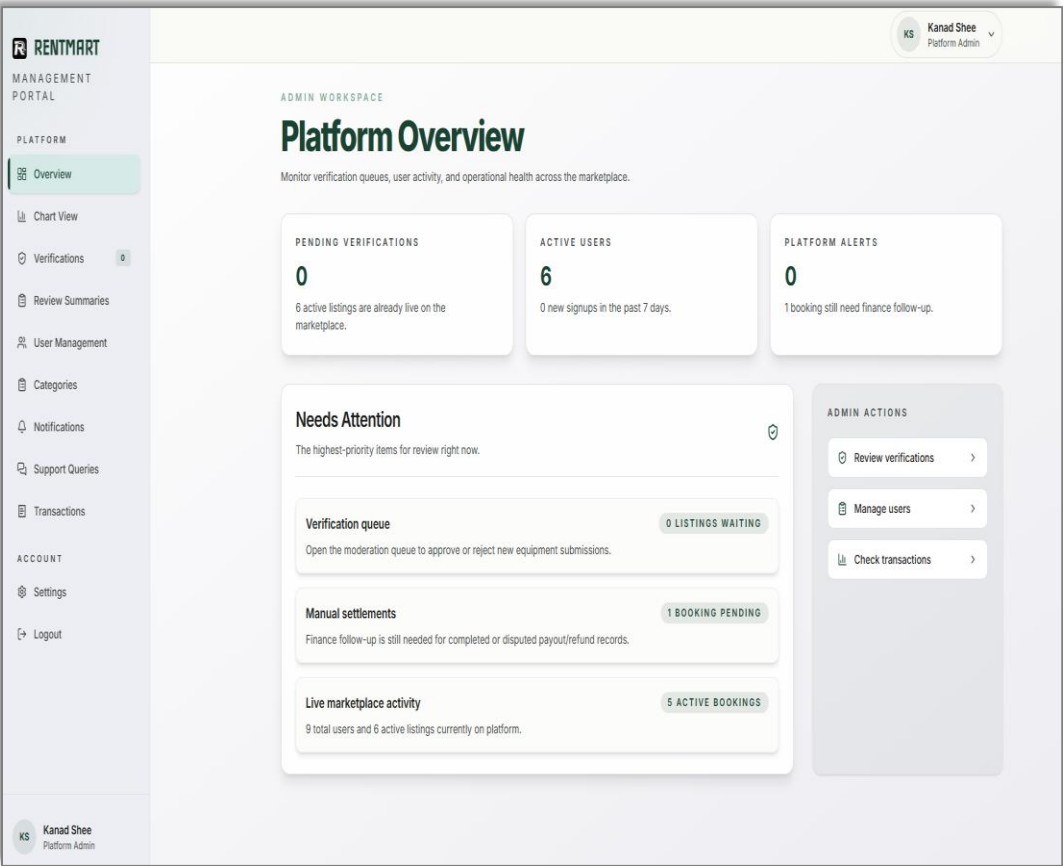


fig: Admin Dashboard page

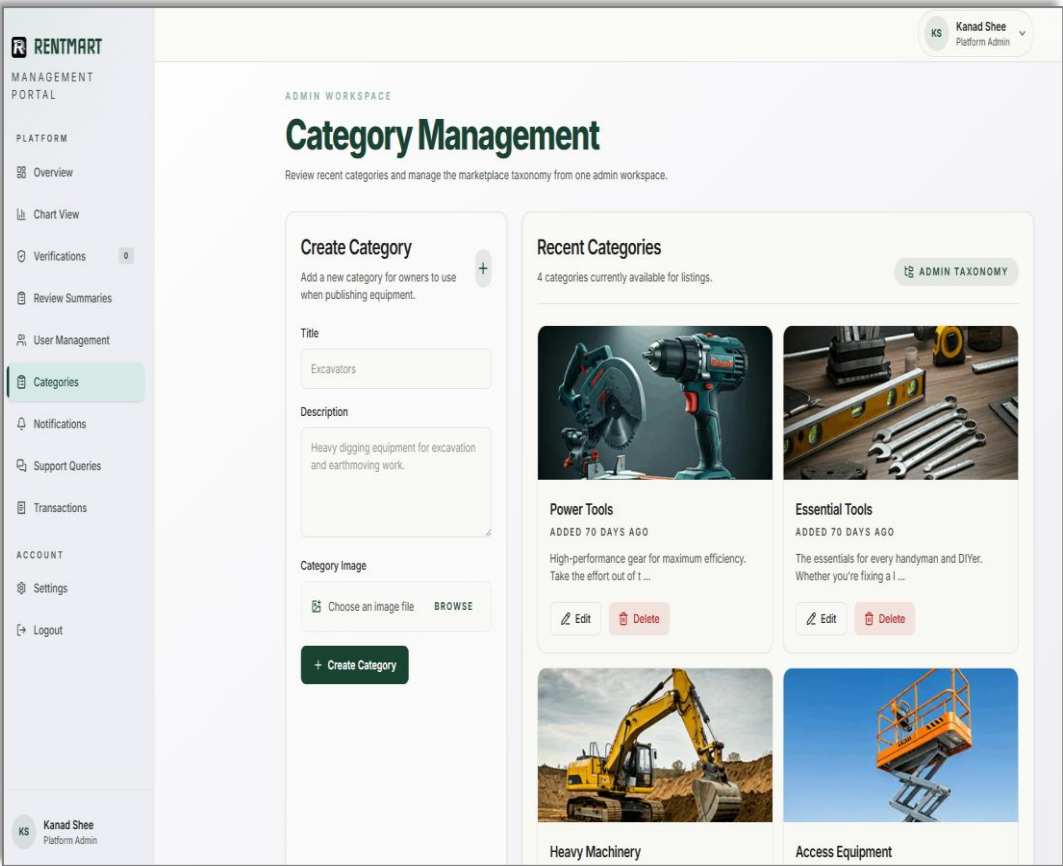


fig: Category management  
page



# OWNER DASHBOARD

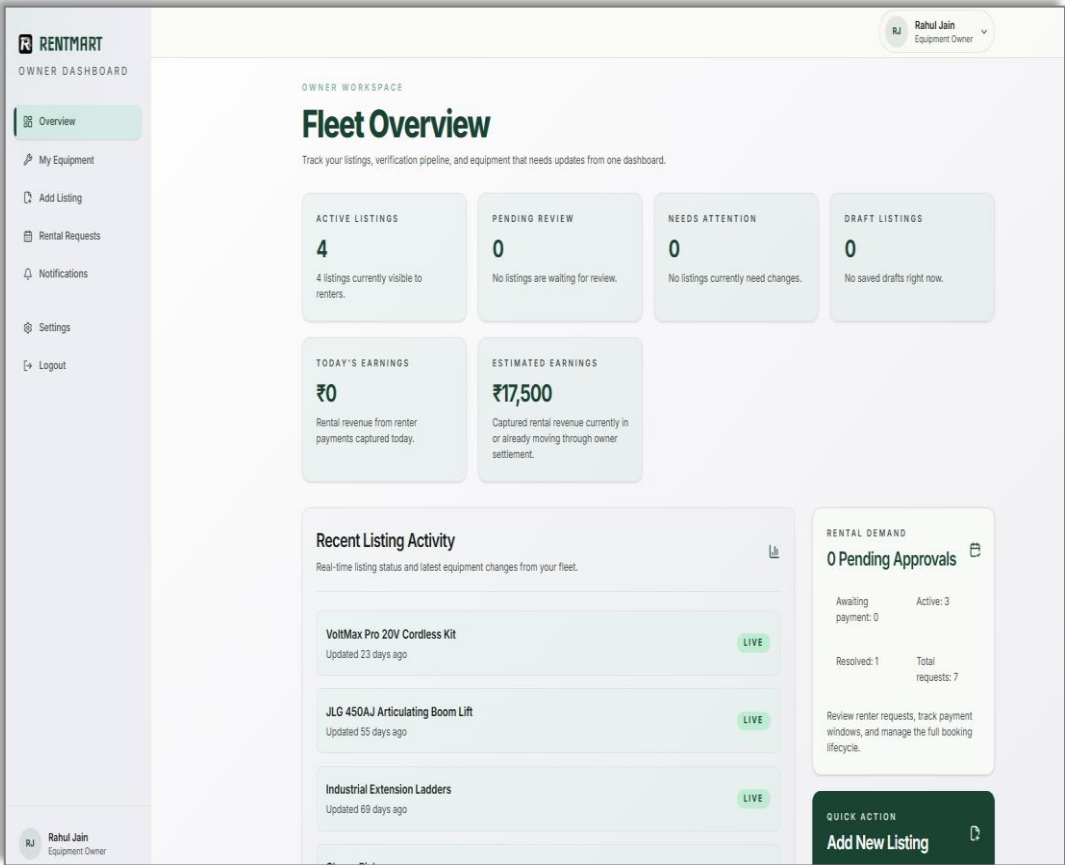


fig: Owner Dashboard page

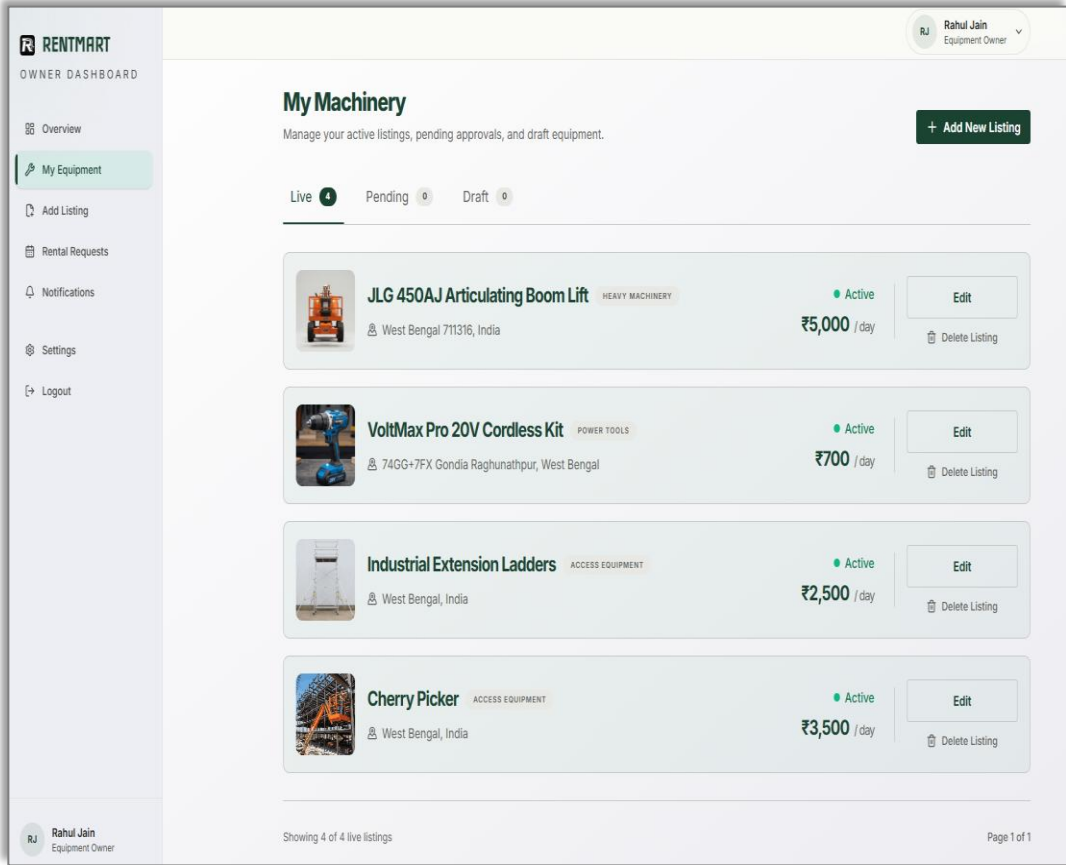


fig : Owner Listing

# RENTER DASHBOARD

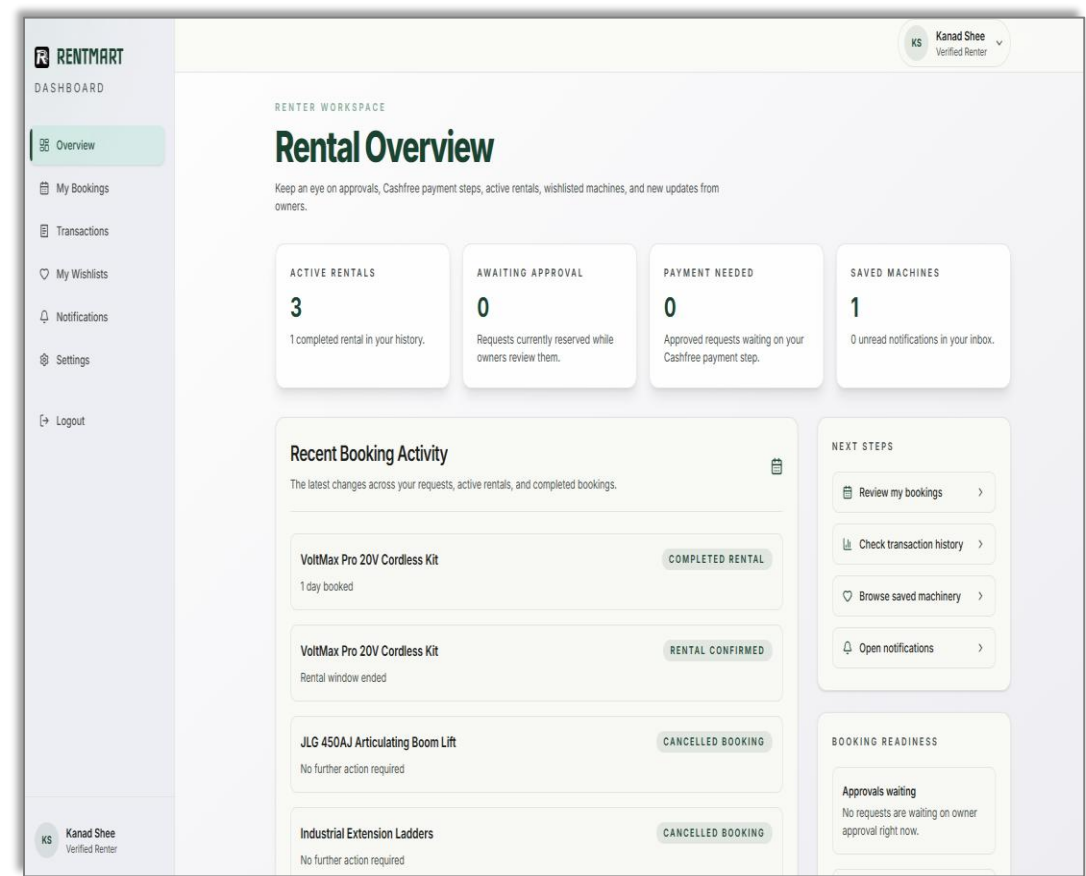


fig: Renter's Dashboard

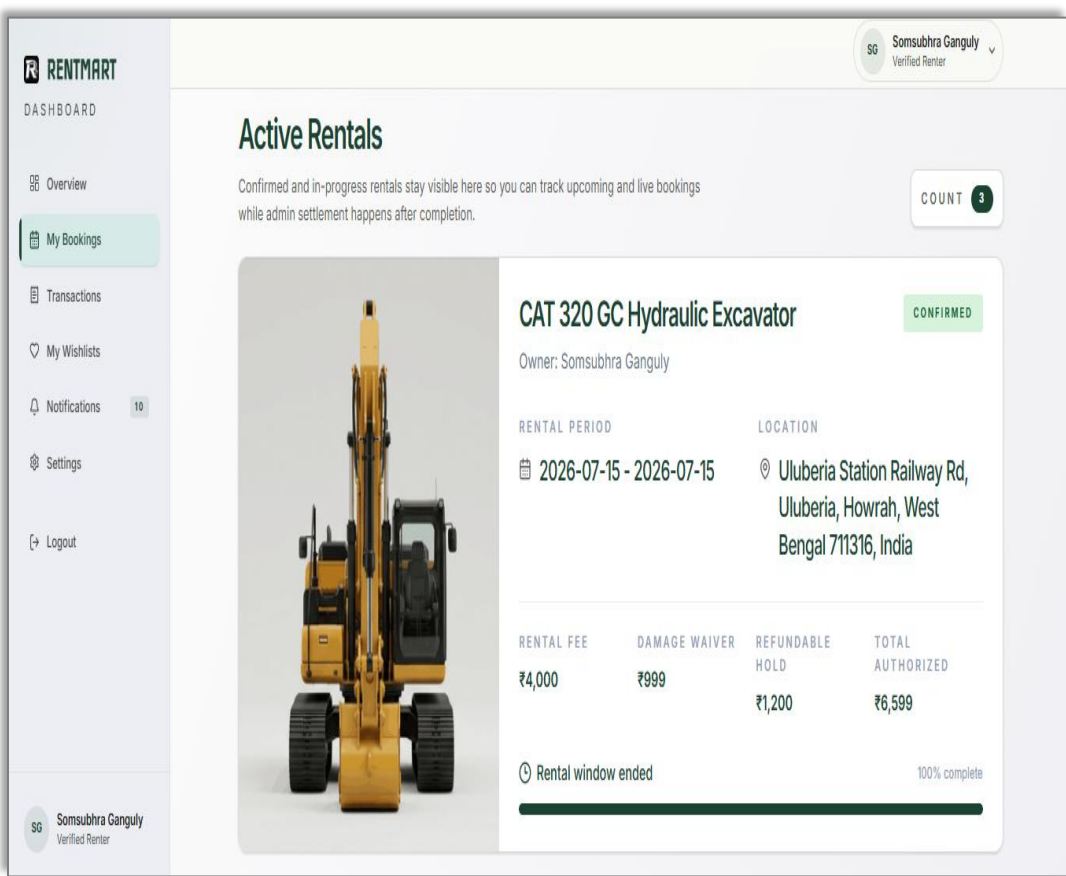


fig: Active rentals page

# CONCLUSION

- **The project provides a simple digital platform to connect equipment owners and renters, making the rental process easier and more organized.**
- **It helps reduce the cost of buying expensive equipment by allowing users to rent them only when needed.**
- **The system also enables owners to earn income from equipment that would otherwise remain unused.**
- **Overall, the platform promotes better resource sharing and improves access to equipment for local and rural communities.**

# LIMITATIONS

- **Currently available only as a web-based platform; a mobile application is planned for future development.**
- **Limited geographic coverage in the initial phase; expansion to more regions will be required.**
- **Internet connectivity is required, which may be a challenge in some rural areas.**
- **Limited equipment categories and users at the beginning, which will grow as the platform scales.**



# FUTURE WORK

- **Development of locality-based search technique to enable optimized recommendations of products.**
- **Scaling the platform to add a new role- user “operator” for heavy machineries which require skilled operator.**
- **AI RAG based model integration for optimized resource planning and analysis purpose.**

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Thank  
You